

**UNIVERSIDAD EL BOSQUE FACULTAD DE INGENIERÍA**



**ASIGNATURA:**

**BASES DE DATOS 1**

**DOCENTE:**

**CHRISTIAN FELIPE DUARTE AREVALO**

**AUTORES:**

**DANIEL FELIPE AGUDELO PEREZ**

**GABRIELA POVEDA GARCIA**

**NATALY MANRIQUE VERA**

**MAYO 2026**

## EVIDENCIAS

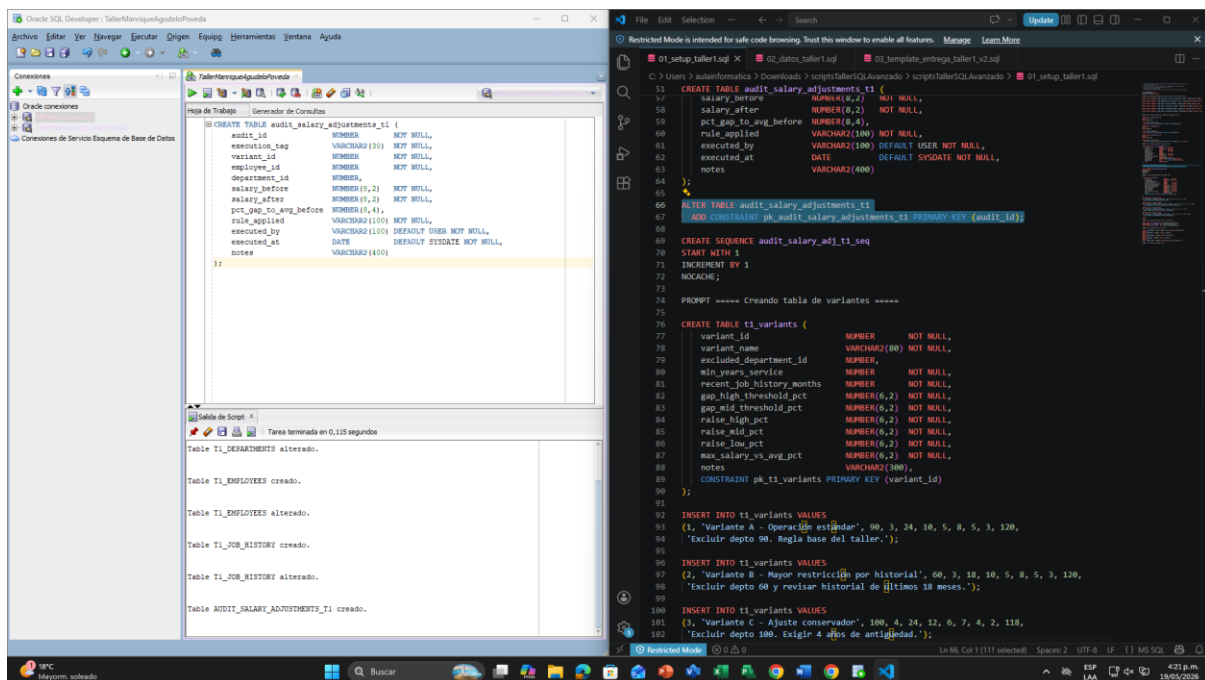
En este taller se trabajó la variante 3:

```
INSERT INTO t1_variants VALUES
```

(3, 'Variante C - Ajuste conservador', 100, 4, 24, 12, 6, 7, 4, 2, 118,

'Excluir depto 100. Exigir 4 años de antigüedad.');

Primero se ingresa toda la información de los scripts a la base de datos con su variante



## PROMP DE LOS DATOS DEL ARCHIVO 02

The screenshot displays the Oracle SQL Developer interface with a SQL script and its execution results.

**SQL Script:**

```
SELECT department_id, department_name
FROM ti_departments
WHERE department_id IN (280, 290)
ORDER BY department_id;
```

**Results:**

| DEPARTMENT_ID | DEPARTMENT_NAME |
|---------------|-----------------|
| 280           | Innovation Lab  |
| 290           | Compliance Cell |

**SQL Script (Continued):**

```
UPDATE ti_employees
SET salary = 6000
WHERE employee_id = 149;

-- 8) Un salario muy cercano al máximo del departamento para forzar validación de tope
UPDATE ti_employees
SET salary = 7700
WHERE employee_id = 174;

COMMIT;

PROMPT ----- Validaciones rápidas del escenario -----

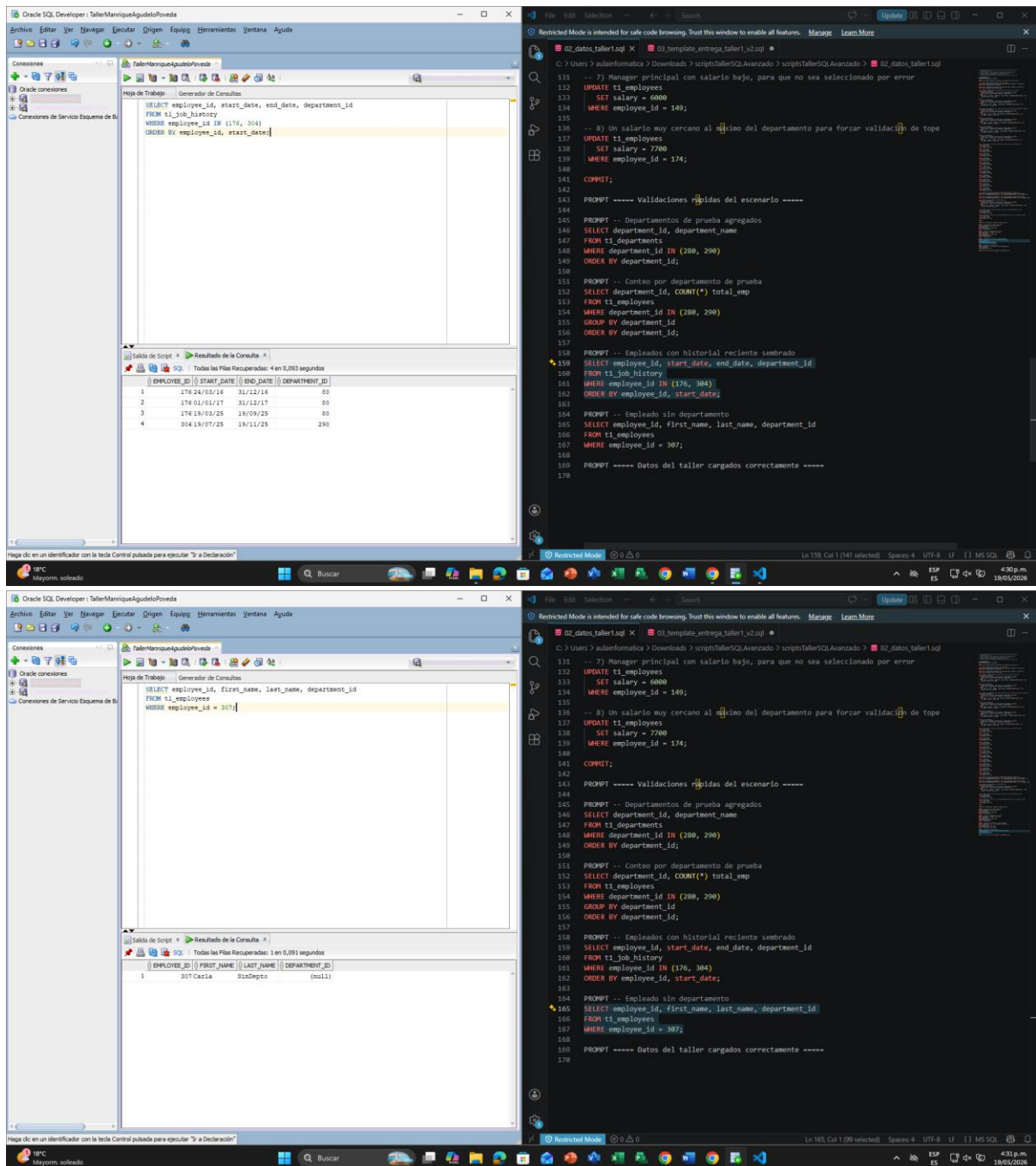
PROMPT -- Departamentos de prueba agregados
SELECT department_id, department_name
FROM ti_departments
WHERE department_id IN (280, 290)
ORDER BY department_id;

PROMPT -- Conteo por departamento de prueba
SELECT department_id, COUNT(*) total_emp
FROM ti_employees
WHERE department_id IN (280, 290)
GROUP BY department_id
ORDER BY department_id;

PROMPT -- Empleados con historial reciente sembrado
SELECT employee_id, start_date, end_date, department_id
FROM ti_job_history
WHERE employee_id IN (176, 304)
ORDER BY employee_id, start_date;

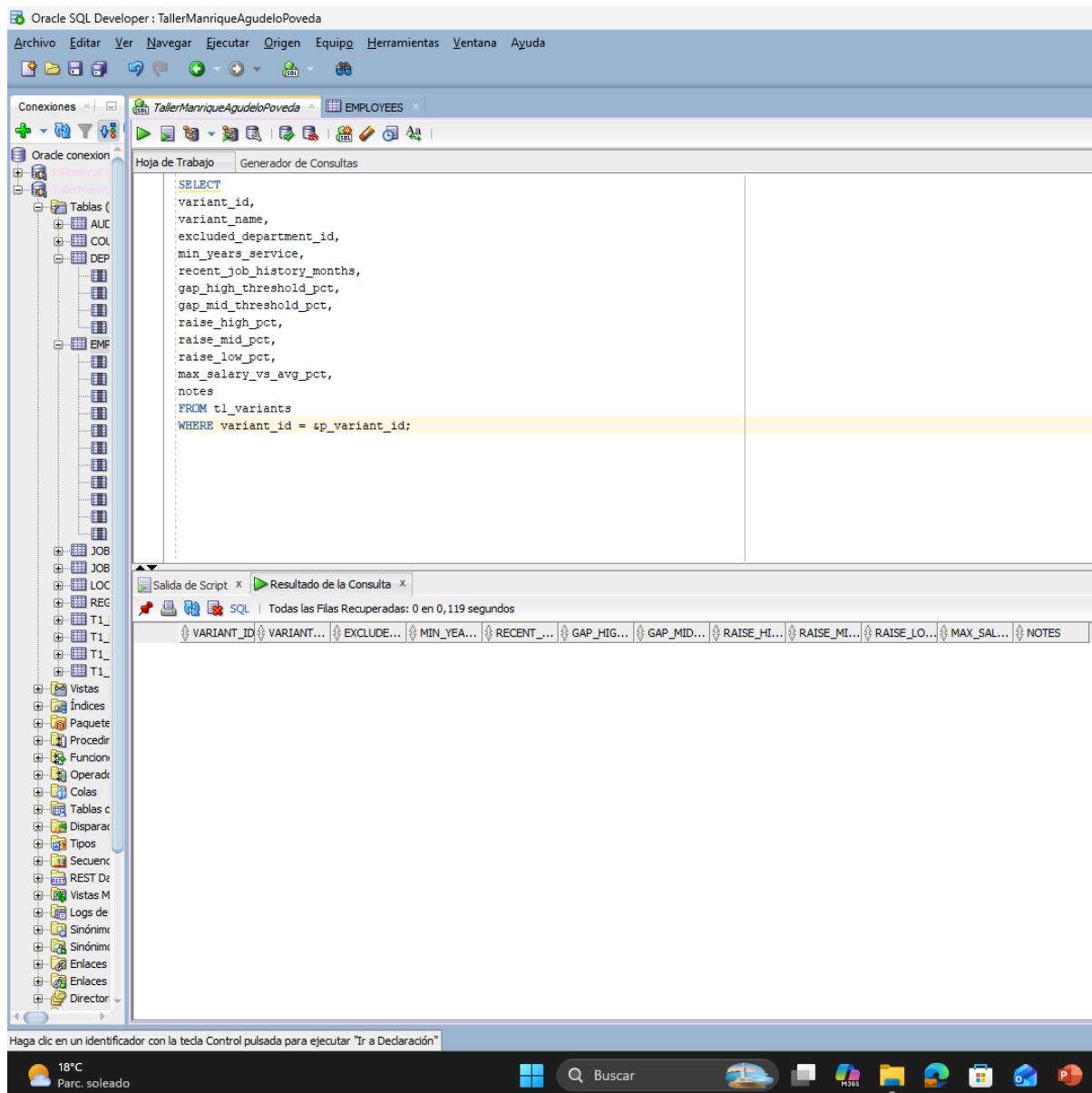
PROMPT -- Empleado sin departamento
SELECT employee_id, first_name, last_name, department_id
FROM ti_employees
WHERE employee_id = 307;

PROMPT ----- Datos del taller cargados correctamente -----
```



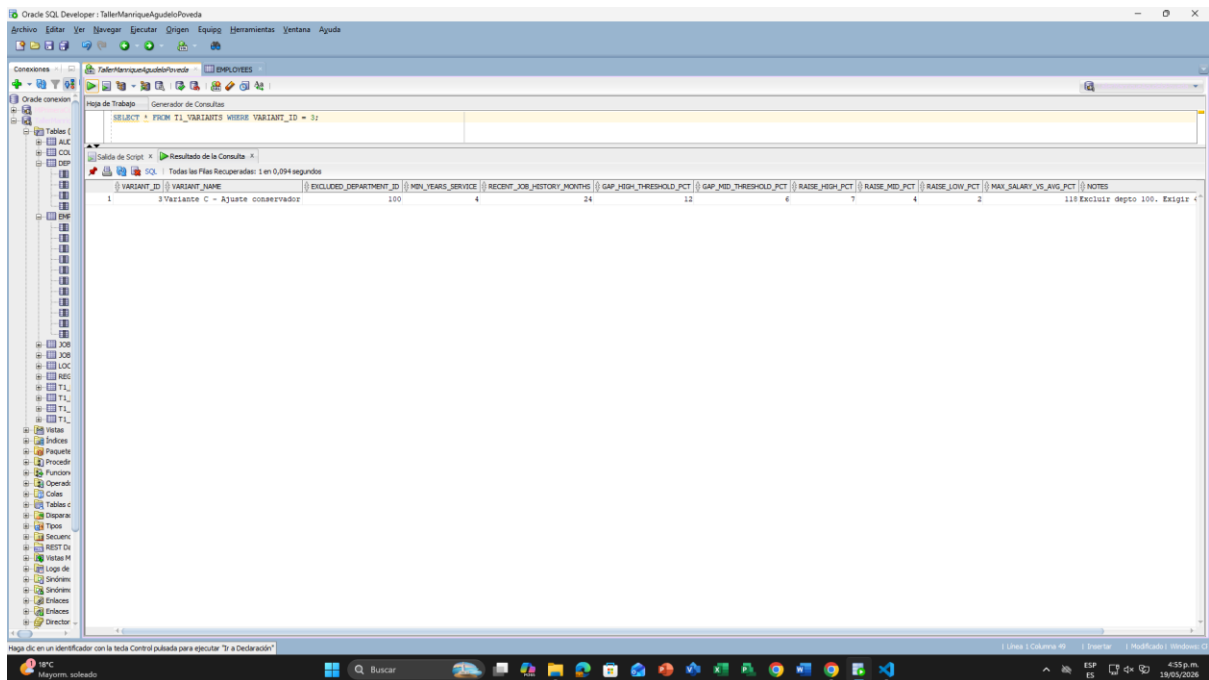
Se empezian a realizar las ejecuciones:

Antes de comenzar se debe realizar esta tabla para poder saber todo sobre la variante y así



### Verificación de la variante

En esta sentencia primero vamos a ver la variante que nos toca en este caso V3, esta consulta lo que nos busca demostrar es todas aquellas variables que necesitamos usar, tener en cuenta y saber para poder ir descartando.



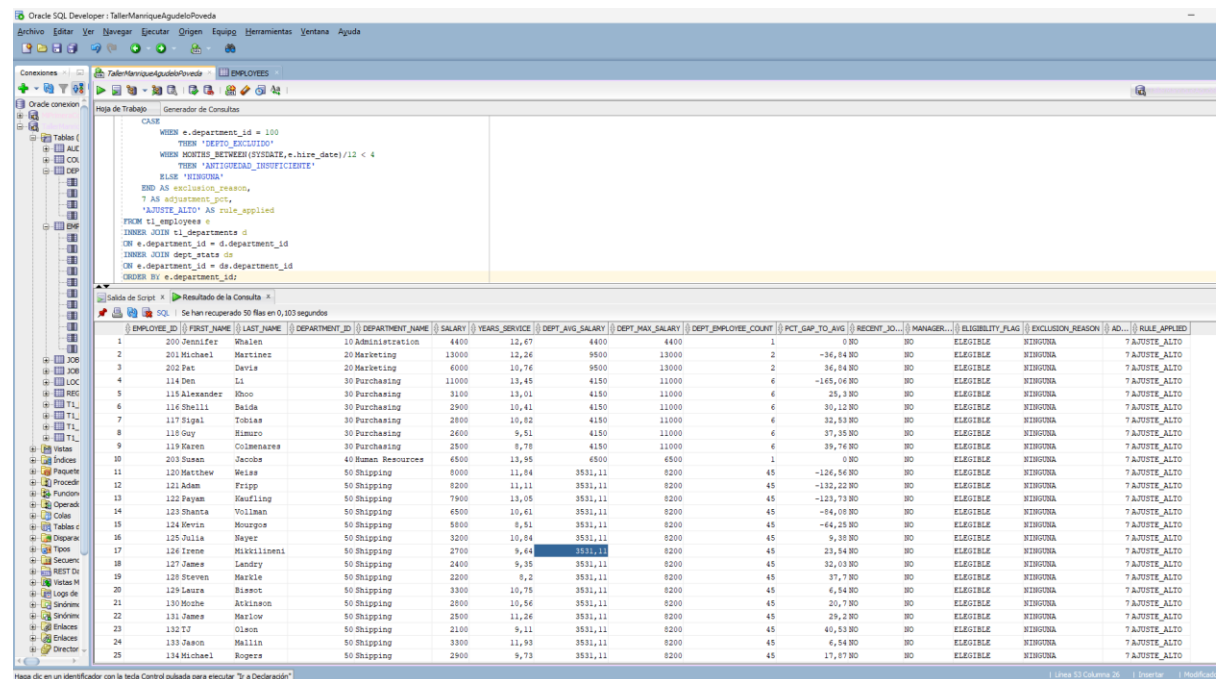
## 1. Consulta diagn stica:

La consulta diagn stica permite analizar salarios, la antigüedad y comportamiento salarial por departamento, tambi n identifica el historial reciente y ranking salarial para la decisi n de empleados elegibles.

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | JOB_ID   | MANAGER_ID | DEPARTMENT_ID | DEPARTMENT_NAME | SALARY | HIRE_DATE | YEARS_SERVICE | DEPT_AVG_SALARY | DEPT_MAX_SALARY | DEPT_EMPLOYEE_COUNT | PCT_GAP_TO_AVG | RECENT_JOB_HISTORY_FLAG | SALARY_RANK_IN_DEPARTMENT |
|-------------|------------|-----------|----------|------------|---------------|-----------------|--------|-----------|---------------|-----------------|-----------------|---------------------|----------------|-------------------------|---------------------------|
| 1           | Dennis     | Wheat     | AD_ASST  | 101        | 10            | Administration  | 4400   | 17/09/13  | 12,47         | 4400            | 4400            | 1                   | 0,00           |                         | 1                         |
| 2           | Michael    | Martinez  | MS_MGR   | 100        | 20            | Marketing       | 13000  | 17/02/14  | 12,24         | 9500            | 13000           | 2                   | -26,84%        |                         | 1                         |
| 3           | Pat        | Davis     | MS_REP   | 201        | 20            | Marketing       | 6000   | 17/09/15  | 10,74         | 9500            | 13000           | 2                   | 36,84%         |                         | 2                         |
| 4           | Den        | Li        | PU_MGR   | 100        | 30            | Purchasing      | 11000  | 07/12/12  | 12,45         | 4150            | 11000           | 6                   | -165,06%       |                         | 1                         |
| 5           | Alexander  | Whoo      | PU_CLERK | 114        | 30            | Purchasing      | 3100   | 18/05/13  | 10,01         | 4150            | 11000           | 6                   | 25,30%         |                         | 2                         |
| 6           | Shelli     | Beale     | PU_CLERK | 114        | 30            | Purchasing      | 2900   | 24/12/15  | 10,41         | 4150            | 11000           | 6                   | 30,12%         |                         | 3                         |
| 7           | Sigal      | Tobias    | PU_CLERK | 114        | 30            | Purchasing      | 2800   | 24/07/15  | 10,52         | 4150            | 11000           | 6                   | 32,53%         |                         | 4                         |
| 8           | Roy        | Kunzo     | PU_CLERK | 114        | 30            | Purchasing      | 2600   | 10/11/14  | 9,51          | 4150            | 11000           | 6                   | 37,35%         |                         | 5                         |
| 9           | Harren     | Colman    | PU_CLERK | 114        | 30            | Purchasing      | 2500   | 10/08/17  | 8,79          | 4150            | 11000           | 6                   | 39,76%         |                         | 6                         |
| 10          | Suan       | Jacobe    | SR_REP   | 101        | 40            | Human Resources | 6500   | 07/06/12  | 11,95         | 6500            | 6500            | 1                   | 0,00           |                         | 1                         |
| 11          | Adam       | Fripp     | ST_MGR   | 100        | 50            | Shipping        | 8200   | 10/04/15  | 11,11         | 3531,11         | 8200            | 45                  | -132,22%       |                         | 1                         |
| 12          | Neena      | Wile      | ST_MGR   | 100        | 50            | Shipping        | 8000   | 10/07/14  | 11,04         | 3531,11         | 8200            | 45                  | -124,56%       |                         | 2                         |
| 13          | Payam      | Kaufling  | ST_MGR   | 100        | 50            | Shipping        | 7900   | 03/05/13  | 10,95         | 3531,11         | 8200            | 45                  | -123,73%       |                         | 3                         |
| 14          | Shanta     | Vollman   | ST_MGR   | 100        | 50            | Shipping        | 6900   | 10/10/15  | 10,41         | 3531,11         | 8200            | 45                  | -84,08%        |                         | 4                         |
| 15          | Kevin      | Mourque   | ST_MGR   | 100        | 50            | Shipping        | 5900   | 14/11/17  | 9,51          | 3531,11         | 8200            | 45                  | -64,25%        |                         | 5                         |
| 16          | Jess       | Fleener   | SH_CLERK | 120        | 50            | Shipping        | 5600   | 23/02/14  | 10,24         | 3531,11         | 8200            | 45                  | -55,59%        |                         | 6                         |
| 17          | Rudica     | Sarchand  | SH_CLERK | 121        | 50            | Shipping        | 4200   | 27/01/14  | 10,31         | 3531,11         | 8200            | 45                  | -18,94%        |                         | 7                         |
| 18          | Alexis     | Beil      | SH_CLERK | 121        | 50            | Shipping        | 4100   | 20/02/15  | 10,29         | 3531,11         | 8200            | 45                  | -16,11%        |                         | 8                         |
| 19          | Sarah      | Beil      | SH_CLERK | 123        | 50            | Shipping        | 4000   | 04/02/15  | 10,29         | 3531,11         | 8200            | 45                  | -13,28%        |                         | 9                         |
| 20          | Britney    | Everett   | SH_CLERK | 123        | 50            | Shipping        | 3900   | 03/03/15  | 11,21         | 3531,11         | 8200            | 45                  | -10,49%        |                         | 10                        |
| 21          | Dell       | Cheng     | SH_CLERK | 123        | 50            | Shipping        | 3800   | 14/04/15  | 10,90         | 3531,11         | 8200            | 45                  | -7,41%         |                         | 11                        |
| 22          | Dennis     | Dilly     | SH_CLERK | 123        | 50            | Shipping        | 3600   | 18/08/15  | 10,77         | 3531,11         | 8200            | 45                  | -1,95%         |                         | 12                        |
| 23          | Bena       | Ladwig    | ST_CLERK | 123        | 50            | Shipping        | 3600   | 14/07/13  | 12,55         | 3531,11         | 8200            | 45                  | -1,95%         |                         | 12                        |
| 24          | Trevor     | Ratz      | ST_CLERK | 124        | 50            | Shipping        | 3500   | 17/10/13  | 12,59         | 3531,11         | 8200            | 45                  | 0,89%          |                         | 13                        |
| 25          | Dalia      | Delinger  | SH_CLERK | 121        | 50            | Shipping        | 3400   | 24/06/16  | 9,91          | 3531,11         | 8200            | 45                  | 5,71%          |                         | 14                        |

## 2. Decisi n de elegibles:

Se aplican las reglas de la variante 3, se excluye el departamento 100 y empleados con menos de 4 años de antigüedad, los que quedaron fueron seleccionados como elegibles



Oracle SQL Developer: TallerManriqueAgudeloPoveda

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Conexiones: TallerManriqueAgudeloPoveda EMPLEADOS

Hoja de Trabajo Generador de Consultas

```
SQL
CASE
  WHEN e.department_id = 100
  THEN 'DEPTO_EXCLUIDO'
  WHEN MONTHS_BETWEEN(SYSDATE, e.hire_date)/12 < 4
  THEN 'ANTIGÜEDAD_INSUFICIENTE'
  ELSE 'NINGUNA'
END AS exclusion_reason,
7 AS adjustment_pct,
'ajuste_salario' AS rule_applied
FROM tl_employees e
INNER JOIN tl_departments d
ON e.department_id = d.department_id
INNER JOIN dept_stats ds
ON e.department_id = ds.department_id
ORDER BY e.department_id
```

Salida de Script X Resultado de la Consulta X

Se han recuperado 50 filas en 0,103 segundos

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | DEPARTMENT_ID | DEPARTMENT_NAME | SALARY | YEARS_SERVICE | DEPT_AVG_SALARY | DEPT_MAX_SALARY | DEPT_EMPLOYEE_COUNT | PCT_GAP_TO_AVG | RECENT_JOB... | MANAGER... | ELIGIBILITY_FLAG | EXCLUSION_REASON | AD... | RULE_APPLIED  |
|-------------|------------|-----------|---------------|-----------------|--------|---------------|-----------------|-----------------|---------------------|----------------|---------------|------------|------------------|------------------|-------|---------------|
| 1           | 200        | Douglas   | 10            | Administration  | 4400   | 12,47         | 4400            | 4400            | 1                   | 0,00           | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 2           | 201        | Michael   | 20            | Marketing       | 13000  | 12,24         | 9500            | 13000           | 2                   | -36,84%        | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 3           | 202        | Rat       | 20            | Marketing       | 6000   | 10,76         | 9500            | 13000           | 2                   | 36,84%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 4           | 114        | Den       | 30            | Purchasing      | 11000  | 13,45         | 4150            | 11000           | 6                   | -165,06%       | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 5           | 115        | Alexander | 30            | Purchasing      | 3100   | 13,01         | 4150            | 11000           | 6                   | 25,30%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 6           | 116        | Shelli    | 30            | Purchasing      | 2900   | 10,41         | 4150            | 11000           | 6                   | 30,12%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 7           | 117        | Stéphane  | 30            | Purchasing      | 2800   | 10,62         | 4150            | 11000           | 6                   | 32,50%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 8           | 118        | Guy       | 30            | Purchasing      | 2400   | 5,51          | 4150            | 11000           | 6                   | 37,35%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 9           | 119        | Karen     | 30            | Purchasing      | 2500   | 8,78          | 4150            | 11000           | 6                   | 39,76%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 10          | 203        | Susan     | 40            | Human Resources | 6500   | 13,95         | 6500            | 6500            | 1                   | 0,00           | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 11          | 120        | Matthew   | 50            | Shipping        | 8000   | 11,04         | 3531,11         | 8200            | 45                  | -124,56%       | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 12          | 121        | Adam      | 50            | Shipping        | 8200   | 11,11         | 3531,11         | 8200            | 45                  | -132,22%       | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 13          | 122        | Peyman    | 50            | Shipping        | 7900   | 13,05         | 3531,11         | 8200            | 45                  | -123,73%       | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 14          | 123        | Shanta    | 50            | Shipping        | 4500   | 10,61         | 3531,11         | 8200            | 45                  | -64,08%        | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 15          | 124        | Kevin     | 50            | Shipping        | 5600   | 8,51          | 3531,11         | 8200            | 45                  | -64,25%        | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 16          | 125        | Tulia     | 50            | Shipping        | 3200   | 10,04         | 3531,11         | 8200            | 45                  | 9,38%          | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 17          | 126        | Irene     | 50            | Shipping        | 2700   | 9,44          | 3531,11         | 8200            | 45                  | 23,84%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 18          | 127        | James     | 50            | Shipping        | 2400   | 9,35          | 3531,11         | 8200            | 45                  | 32,03%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 19          | 128        | Steven    | 50            | Shipping        | 2200   | 8,2           | 3531,11         | 8200            | 45                  | 37,70%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 20          | 129        | Laura     | 50            | Shipping        | 3300   | 10,75         | 3531,11         | 8200            | 45                  | 6,54%          | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 21          | 130        | Mozhe     | 50            | Shipping        | 2800   | 10,56         | 3531,11         | 8200            | 45                  | 20,70%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 22          | 131        | James     | 50            | Shipping        | 2500   | 11,24         | 3531,11         | 8200            | 45                  | 29,20%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 23          | 132        | TJ        | 50            | Shipping        | 2100   | 9,11          | 3531,11         | 8200            | 45                  | 40,50%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 24          | 133        | Daan      | 50            | Shipping        | 3300   | 11,93         | 3531,11         | 8200            | 45                  | 6,54%          | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |
| 25          | 134        | Michael   | 50            | Shipping        | 2900   | 9,73          | 3531,11         | 8200            | 45                  | 17,87%         | NO            | NO         | ELEGIBLE         | NINGUNA          |       | 7 AJUSTE_ALTO |

Hoja clic en un identificador con la tecla Control pulsada para ejecutar "Si se Declaraación"

1 Línea 53 Columna 25 | Insertar | Modificar

3. Prevalidación: Verifica los empleados elegibles para el ajuste salarial y calcula el impacto económico del aumento, también muestra el detalle de los salarios antes y después del ajuste y valida los límites salariales por departamento. Este proceso ayuda a evitar errores antes de aplicar la actualización.

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Conexiones x

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TallerManriqueAgudeloPoveda x EMPLOYEES x

Hoja de Trabajo Generador de Consultas

```

SELECT
COUNT(*) total_eligible_employees,
SUM(salary) total_salary_before,
SUM(salary * 1.07) total_salary_after,
SUM((salary * 1.07) - salary)
AS total_increment
FROM tl_employees
WHERE department_id <> 100
AND MONTHS_BETWEEN(SYSDATE,hire_date)/12 >= 4;

```

Salida de Script x Resultado de la Consulta x

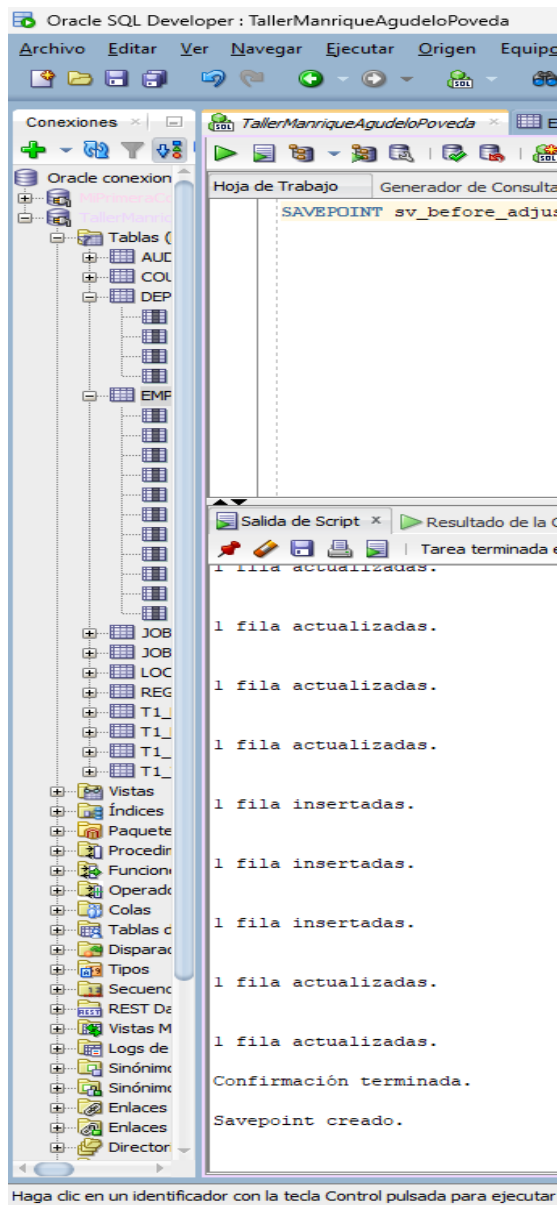
Todas las Filas Recuperadas: 1 en 0,094 segundos

|   | TOTAL_ELIGIBLE_EMPLOYEES | TOTAL_SALARY_BEFORE | TOTAL_SALARY_AFTER | TOTAL_INCREMENT |
|---|--------------------------|---------------------|--------------------|-----------------|
| 1 | 105                      | 644908              | 690051,56          | 45143,56        |

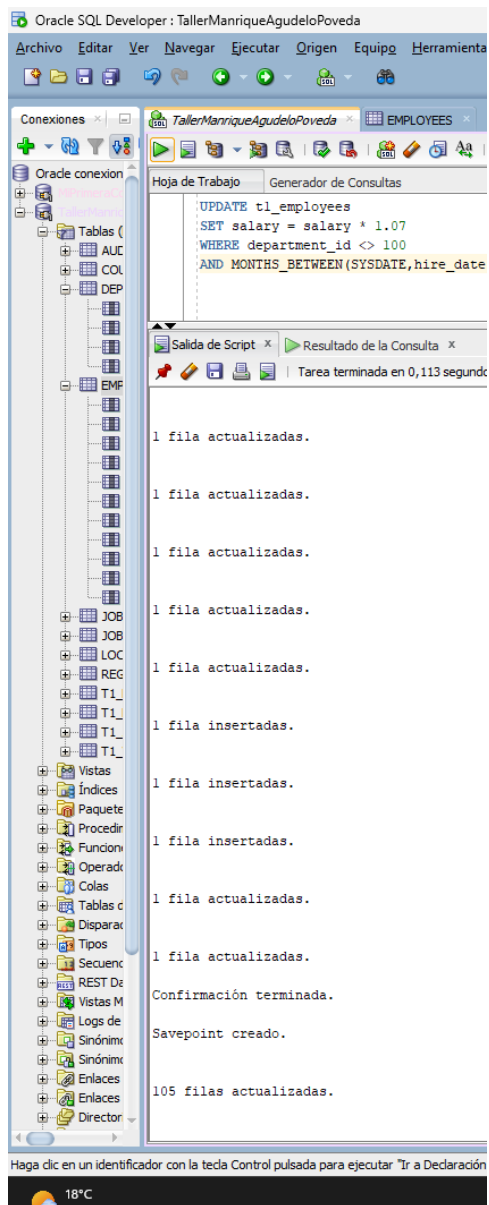
#### 4. Ejecución transaccional:

Se realizo commit porque ahora los datos actualizados cumplen con las condiciones de las variantes y de los topes que se establecieron en los salarios de los empleados

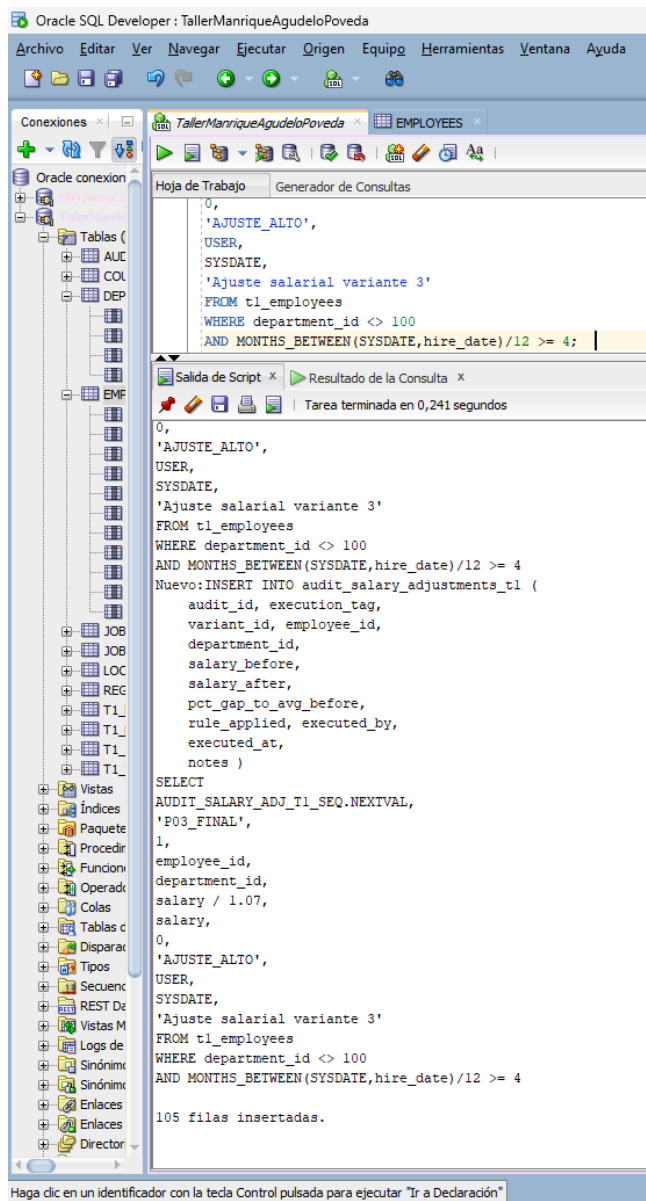




4.1. Actualización de salarios: Se hace la actualización del salario de los empleados que fueron elegibles



4.2. Inserción en auditoria: Se ingresa a la auditoria la informacion de los empleados seleccionados.



### 4.3. Validación intermedia

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Conexiones x TallerManriqueAgudeloPoveda EMPLOYEES

Hoja de Trabajo Generador de Consultas

```

ROUND(salary / 1.07,2)
AS original_salary,
ROUND(
  AVG(salary)
  OVER(PARTITION BY department_id) * 1.18, 2
) AS allowed_max_salary,
'CUMPLE' AS validation_status
FROM tl_employees;

```

Salida de Script x Resultado de la Consulta x

SQL | Se han recuperado 50 filas en 0,097 segundos

| EMPLOYEE_ID | DEPARTMENT_ID | CURRENT_SALARY | ORIGINAL_SALARY | ALLOWED_MAX_SALARY | VALIDATION_STATUS |
|-------------|---------------|----------------|-----------------|--------------------|-------------------|
| 1           | 200           | 10             | 4708            | 4400               | 5555,44 CUMPLE    |
| 2           | 201           | 20             | 13910           | 13000              | 11994,7 CUMPLE    |
| 3           | 202           | 20             | 6420            | 6000               | 11994,7 CUMPLE    |
| 4           | 114           | 30             | 11770           | 11000              | 5239,79 CUMPLE    |
| 5           | 115           | 30             | 3317            | 3100               | 5239,79 CUMPLE    |
| 6           | 116           | 30             | 3103            | 2900               | 5239,79 CUMPLE    |
| 7           | 117           | 30             | 2996            | 2800               | 5239,79 CUMPLE    |
| 8           | 118           | 30             | 2782            | 2600               | 5239,79 CUMPLE    |
| 9           | 119           | 30             | 2675            | 2500               | 5239,79 CUMPLE    |
| 10          | 203           | 40             | 6955            | 6500               | 8206,9 CUMPLE     |
| 11          | 120           | 50             | 8560            | 8000               | 4458,38 CUMPLE    |
| 12          | 121           | 50             | 8774            | 8200               | 4458,38 CUMPLE    |
| 13          | 122           | 50             | 8453            | 7900               | 4458,38 CUMPLE    |
| 14          | 123           | 50             | 6955            | 6500               | 4458,38 CUMPLE    |
| 15          | 124           | 50             | 6206            | 5800               | 4458,38 CUMPLE    |
| 16          | 125           | 50             | 3424            | 3200               | 4458,38 CUMPLE    |
| 17          | 126           | 50             | 2889            | 2700               | 4458,38 CUMPLE    |
| 18          | 127           | 50             | 2568            | 2400               | 4458,38 CUMPLE    |
| 19          | 128           | 50             | 2354            | 2200               | 4458,38 CUMPLE    |
| 20          | 129           | 50             | 3531            | 3300               | 4458,38 CUMPLE    |
| 21          | 130           | 50             | 2996            | 2800               | 4458,38 CUMPLE    |
| 22          | 131           | 50             | 2675            | 2500               | 4458,38 CUMPLE    |
| 23          | 132           | 50             | 2247            | 2100               | 4458,38 CUMPLE    |
| 24          | 133           | 50             | 3531            | 3300               | 4458,38 CUMPLE    |
| 25          | 134           | 50             | 3103            | 2900               | 4458,38 CUMPLE    |
| 26          | 135           | 50             | 2568            | 2400               | 4458,38 CUMPLE    |
| 27          | 136           | 50             | 2354            | 2200               | 4458,38 CUMPLE    |
| 28          | 137           | 50             | 3852            | 3600               | 4458,38 CUMPLE    |
| 29          | 138           | 50             | 3424            | 3200               | 4458,38 CUMPLE    |
| 30          | 139           | 50             | 2889            | 2700               | 4458,38 CUMPLE    |
| 31          | 140           | 50             | 2675            | 2500               | 4458,38 CUMPLE    |

Una de las identidades con la que Control se debe crear al crear "Tr = Transacción"

#### 4.4. Control transaccional

#### 5. Validación posterior

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Conexiones x TallerManriqueAgudeloPoveda x EMPLOYEES x

Hoja de Trabajo Generador de Consultas

```
SELECT
employee_id,
department_id,
salary_before,
salary_after,
execution_tag
FROM audit_salary_adjustments_t1
WHERE execution_tag = 'sp_execution_tag';
```

Salida de Script x Resultado de la Consulta x

SQL | Se han recuperado 50 filas en 0,101 segundos

|    | EMPLOYEE_ID | DEPARTMENT_ID | SALARY_BEFORE | SALARY_AFTER | EXECUTION_TAG |
|----|-------------|---------------|---------------|--------------|---------------|
| 1  | 100         | 90            | 24000         | 25680        | P03_FINAL     |
| 2  | 101         | 90            | 17000         | 18190        | P03_FINAL     |
| 3  | 102         | 90            | 17000         | 18190        | P03_FINAL     |
| 4  | 103         | 60            | 9000          | 9630         | P03_FINAL     |
| 5  | 104         | 60            | 6000          | 6420         | P03_FINAL     |
| 6  | 105         | 60            | 4800          | 5136         | P03_FINAL     |
| 7  | 106         | 60            | 4800          | 5136         | P03_FINAL     |
| 8  | 107         | 60            | 4200          | 4494         | P03_FINAL     |
| 9  | 114         | 30            | 11000         | 11770        | P03_FINAL     |
| 10 | 115         | 30            | 3100          | 3317         | P03_FINAL     |
| 11 | 116         | 30            | 2900          | 3103         | P03_FINAL     |
| 12 | 117         | 30            | 2800          | 2996         | P03_FINAL     |
| 13 | 118         | 30            | 2600          | 2782         | P03_FINAL     |
| 14 | 119         | 30            | 2500          | 2675         | P03_FINAL     |
| 15 | 120         | 50            | 8000          | 8560         | P03_FINAL     |
| 16 | 121         | 50            | 8200          | 8774         | P03_FINAL     |
| 17 | 122         | 50            | 7900          | 8453         | P03_FINAL     |
| 18 | 123         | 50            | 6500          | 6955         | P03_FINAL     |
| 19 | 124         | 50            | 5800          | 6206         | P03_FINAL     |
| 20 | 125         | 50            | 3200          | 3424         | P03_FINAL     |
| 21 | 126         | 50            | 2700          | 2889         | P03_FINAL     |
| 22 | 127         | 50            | 2400          | 2568         | P03_FINAL     |
| 23 | 128         | 50            | 2200          | 2354         | P03_FINAL     |
| 24 | 129         | 50            | 3300          | 3531         | P03_FINAL     |
| 25 | 130         | 50            | 2800          | 2996         | P03_FINAL     |
| 26 | 131         | 50            | 2500          | 2675         | P03_FINAL     |
| 27 | 132         | 50            | 2100          | 2247         | P03_FINAL     |
| 28 | 133         | 50            | 3300          | 3531         | P03_FINAL     |
| 29 | 134         | 50            | 2900          | 3103         | P03_FINAL     |
| 30 | 135         | 50            | 2400          | 2568         | P03_FINAL     |
| 31 | 136         | 50            | 2200          | 2354         | P03_FINAL     |

Haga clic en un identificador con la tecla Control pulsada para ejecutar "Ir a Declaración"

17°C Parc. soleado

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Conexiones x TallerManriqueAgudeloPoveda x EMPLOYEES x

Hoja de Trabajo Generador de Consultas

```
SUM(salary_before)
total_salary_before,
SUM(salary_after)
total_salary_after,
SUM(salary_after - salary_before)
total_increment
FROM audit_salary_adjustments_t1
WHERE execution_tag = 'sp_execution_tag';
```

Salida de Script x Resultado de la Consulta x

SQL | Todas las Filas Recuperadas: 1 en 0,094 segundos

|   | TOTAL_ROWS_AUDITED | TOTAL_SALARY_BEFORE | TOTAL_SALARY_AFTER | TOTAL_INCREMENT |
|---|--------------------|---------------------|--------------------|-----------------|
| 1 | 105                | 644908              | 690051,56          | 45143,56        |

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Conexiones

Oracle conexiones

Tablas (Filtrado)

- AUDIT\_SALARY\_ADJUSTMENTS\_T1
- COUNTRIES
- DEPARTMENTS
  - DEPARTMENT\_ID
  - DEPARTMENT\_NAME
  - MANAGER\_ID
  - LOCATION\_ID
- EMPLOYEES
  - EMPLOYEE\_ID
  - FIRST\_NAME
  - LAST\_NAME
  - EMAIL
  - PHONE\_NUMBER
  - HIRE\_DATE
  - JOB\_ID
  - SALARY
  - COMMISSION\_PCT
  - MANAGER\_ID
  - DEPARTMENT\_ID
- JOB\_HISTORY
- JOBS
- LOCATIONS
- REGIONS
- T1\_DEPARTMENTS
- T1\_EMPLOYEES
- T1\_JOB\_HISTORY
- T1\_VARIANTS

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Colas

Tablas de Colas

Disparadores

Tipos

Secuencias

REST Data Services

TallerManriqueAgudeloPoveda EMPLOYEES

Hoja de Trabajo Generador de Consultas

```
SELECT
employee_id,
department_id,
salary_after,
ROUND(
    AVG(salary_after)
    OVER(PARTITION BY department_id) * 1.18, 2
) AS allowed_max_salary,
'VALIDO' AS top_limit_status
FROM audit_salary_adjustments_t1
WHERE execution_tag = 'sp_execution_tag';
```

Salida de Script x Resultado de la Consulta x

SQL | Se han recuperado 50 filas en 0,095 segundos

|    | EMPLOYEE_ID | DEPARTMENT_ID | SALARY_AFTER | ALLOWED_MAX_SALARY | TOP_LIMIT_STATUS |
|----|-------------|---------------|--------------|--------------------|------------------|
| 1  | 200         | 10            | 4708         | 5555,44            | VALIDO           |
| 2  | 201         | 20            | 13910        | 11994,7            | VALIDO           |
| 3  | 202         | 20            | 6420         | 11994,7            | VALIDO           |
| 4  | 114         | 30            | 11770        | 5239,79            | VALIDO           |
| 5  | 115         | 30            | 3317         | 5239,79            | VALIDO           |
| 6  | 116         | 30            | 3103         | 5239,79            | VALIDO           |
| 7  | 117         | 30            | 2996         | 5239,79            | VALIDO           |
| 8  | 118         | 30            | 2782         | 5239,79            | VALIDO           |
| 9  | 119         | 30            | 2675         | 5239,79            | VALIDO           |
| 10 | 203         | 40            | 6955         | 8206,9             | VALIDO           |
| 11 | 120         | 50            | 8560         | 4458,38            | VALIDO           |
| 12 | 121         | 50            | 8774         | 4458,38            | VALIDO           |
| 13 | 122         | 50            | 8453         | 4458,38            | VALIDO           |
| 14 | 123         | 50            | 6955         | 4458,38            | VALIDO           |
| 15 | 124         | 50            | 6206         | 4458,38            | VALIDO           |
| 16 | 125         | 50            | 3424         | 4458,38            | VALIDO           |
| 17 | 126         | 50            | 2889         | 4458,38            | VALIDO           |
| 18 | 127         | 50            | 2568         | 4458,38            | VALIDO           |
| 19 | 128         | 50            | 2354         | 4458,38            | VALIDO           |
| 20 | 129         | 50            | 3531         | 4458,38            | VALIDO           |

Oracle SQL Developer: TallerManriqueAgudeloPoveda

Archivo Editar Ver Navegar Ejecutar Origen Equipos Herramientas Ventana Ayuda

Conexiones

Oracle conexiones

TallerManriqueAgudeloPoveda EMPLOYEES

Hoja de Trabajo Generador de Consultas

```

SELECT
audit_id,
execution_tag,
variant_id,
employee_id,
department_id,
salary_before,
salary_after,
rule_applied,
executed_by,
executed_at
FROM audit_salary_adjustments_t1
WHERE execution_tag = 'sp_execution_tag';

```

Salida de Script x Resultado de la Consulta x

Se han recuperado 50 filas en 0,1 segundos

|    | AUDIT_ID | EXECUTION_TAG | VARIANT_ID | EMPLOYEE_ID | DEPARTMENT_ID | SALARY_BEFORE | SALARY_AFTER | RULE_APPLIED | EXECUTED_BY | EXECUTED_AT |
|----|----------|---------------|------------|-------------|---------------|---------------|--------------|--------------|-------------|-------------|
| 1  | 1        | P03_FINAL     | 1          | 100         | 90            | 24000         | 25680        | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 2  | 2        | P03_FINAL     | 1          | 101         | 90            | 17000         | 18190        | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 3  | 3        | P03_FINAL     | 1          | 102         | 90            | 17000         | 18190        | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 4  | 4        | P03_FINAL     | 1          | 103         | 60            | 9000          | 9630         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 5  | 5        | P03_FINAL     | 1          | 104         | 60            | 6000          | 6420         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 6  | 6        | P03_FINAL     | 1          | 105         | 60            | 4800          | 5136         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 7  | 7        | P03_FINAL     | 1          | 106         | 60            | 4800          | 5136         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 8  | 8        | P03_FINAL     | 1          | 107         | 60            | 4200          | 4494         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 9  | 9        | P03_FINAL     | 1          | 114         | 30            | 11000         | 11770        | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 10 | 10       | P03_FINAL     | 1          | 115         | 30            | 3100          | 3317         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 11 | 11       | P03_FINAL     | 1          | 116         | 30            | 2900          | 3103         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 12 | 12       | P03_FINAL     | 1          | 117         | 30            | 2800          | 2996         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 13 | 13       | P03_FINAL     | 1          | 118         | 30            | 2600          | 2782         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 14 | 14       | P03_FINAL     | 1          | 119         | 30            | 2500          | 2675         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 15 | 15       | P03_FINAL     | 1          | 120         | 50            | 8000          | 8560         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 16 | 16       | P03_FINAL     | 1          | 121         | 50            | 8200          | 8774         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 17 | 17       | P03_FINAL     | 1          | 122         | 50            | 7900          | 8453         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 18 | 18       | P03_FINAL     | 1          | 123         | 50            | 6500          | 6955         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 19 | 19       | P03_FINAL     | 1          | 124         | 50            | 5800          | 6206         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 20 | 20       | P03_FINAL     | 1          | 125         | 50            | 3200          | 3424         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 21 | 21       | P03_FINAL     | 1          | 126         | 50            | 2700          | 2889         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 22 | 22       | P03_FINAL     | 1          | 127         | 50            | 2400          | 2568         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 23 | 23       | P03_FINAL     | 1          | 128         | 50            | 2200          | 2354         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 24 | 24       | P03_FINAL     | 1          | 129         | 50            | 3300          | 3531         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 25 | 25       | P03_FINAL     | 1          | 130         | 50            | 2800          | 2996         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |
| 26 | 26       | P03_FINAL     | 1          | 131         | 50            | 2500          | 2675         | AJUSTE_ALTO  | GABRIELA    | 19/05/26    |

Haga clic en un identificador con la tecla Control pulsada para ejecutar "Ir a Declaración"

## 7. Justificación técnica

- **ATOMICIDAD:** La transacción asegura que todos los cambios se ejecuten completamente o no se ejecuten.
- **CONSISTENCIA:** Los datos se mantienen válidos porque solo se actualizaron empleados elegibles.
- **AISLAMIENTO:** Oracle controla que otras sesiones no vean cambios sin confirmar durante la transacción.
- **DURABILIDAD:** Después del COMMIT los cambios quedan almacenados permanentemente en la base de datos.
- **USO DE SAVEPOINT/ROLLBACK:** El SAVEPOINT permite regresar al estado anterior si ocurre algún error en la actualización.